
Calibration-Driven Sparse Attention Fine-Tuning (C-SAF) in LLMs

Khalid Shaikh¹ Shikhar Shiromani² Rachit Chadha¹ Azeez Ishaqui¹

Abstract

Deploying large language models (LLMs) with specialized adapters for multiple tasks in resource-constrained environments, such as mobile devices - poses significant hurdles in terms of memory usage, loading overhead, and runtime latency, specially when the number of tasks increase. Existing parameter-efficient fine-tuning (PEFT) methods, such as LoRA, often assume uniform layer importance and finetune all layers of the model, leading to redundant parameters. We propose Calibration-Driven Sparse Attention Fine-Tuning (C-SAF), a novel framework that selectively fine-tunes only the non-complementary layers that critically impact task performance. By focusing LoRA updates on just a small subset of layers, C-SAF cuts adapter sizes by 75%, dramatically reducing both cold-start time and memory usage, making on-device ML workflows - like Apple’s, more efficient. Furthermore, C-SAF achieves up to a 23% speedup in inference for generative tasks, supporting rapid specialization in dynamic, multi-task settings. Across diverse benchmarks and model architectures, our extensive evaluations confirm that C-SAF outperforms existing approaches and sets a new standard for parameter-efficient LLM adaptation under tight resource constraints.

1. Introduction

Transfer learning, specifically fine-tuning large language models (LLMs) to new downstream tasks has become a mainstay of modern natural language processing pipelines. However, as model scales continue to increase—often reach-

ing tens or even hundreds of billions of parameters—fine-tuning all model layers leads to high number of trainable parameters, computation costs and extended training times. Conventional parameter-efficient fine-tuning (PEFT) methods, such as LoRA, have mitigated some of these burdens by updating only a small fraction of parameters. Still, these approaches often apply uniform adjustments across all layers, treating them as equally essential for the target task. This assumption can result in unnecessary overhead, bloated adapter sizes, and missed opportunities for performance and latency improvements.

In this work, we introduce **Calibration-Driven Sparse Attention Fine-Tuning (C-SAF)**, a novel framework that addresses these challenges head-on. C-SAF systematically determines a small subset of transformer layers that meaningfully contribute to task performance by analyzing their behavior after a specialized calibration process. This process leverages attention distributions—particularly the disproportionate focus on certain “attention sink” tokens—to identify layers whose re-calibrated attention patterns lead to measurable performance gains. By categorizing layers as either *complementary* or *non-complementary* through this analysis, C-SAF pinpoints exactly which subset of non-complementary layers that needs fine-tuning, thereby reducing the LoRA adapter size by over 75% while increasing/maintaining the accuracy of LLMs.

Taken together, C-SAF provides a path to more agile, resource-efficient, and performance-optimized fine-tuning, setting a new standard for deploying LLMs in both scalable data centers and memory-constrained edge environments, extending the reach of LLMs to environments where memory and compute are at a premium.

2. Related Work

Training large pre-trained language models, such as GPT-3 with 175 billion parameters, incurs significant computational costs due to their substantial memory and processing requirements. This has contributed to the widespread adoption of transfer learning techniques that utilize open source models, allowing their adaptation to specialized tasks without the need for end-to-end training. Among these techniques, fine-tuning has emerged as a powerful approach to adapt accordingly. As machine learning systems are

¹School of Computer Science, Georgia Institute of Technology, Atlanta, USA ²School of Computational Science and Engineering, Georgia Institute of Technology, Atlanta, USA. Correspondence to: Khalid Shaikh <kshaikh6@gatech.edu>, Shikhar Shiromani <sshiromani3@gatech.edu>, Rachit Chadha <rchadha33@gatech.edu>, Azeez Ishaqui <aishaqui3@gatech.edu>.

continuously deployed at scale across organizations, optimizing fine-tuning methodologies have become critical. In particular, parameter-efficient fine-tuning (PEFT) methods, such as Low-Rank Adaptation (LoRA), have gained prominence by enabling task-specific adaptation while leveraging only a small subset of parameters. This section provides an overview of advancements in PEFT techniques, with a focus on LoRA, examines their efficiency and limitations, and outlines how these insights motivate the optimizations proposed in this work based on the limitations of the attention mechanism.

2.1. Parameter-Efficient Fine-Tuning Methods

Low-Rank Adaptation (LoRA) (Hu et al., 2021) is a foundational PEFT method that addresses the inefficiency of full model fine-tuning. LoRA introduces trainable low-rank matrices into the transformer layers of a pre-trained model while keeping the original weights frozen. Specifically, it injects rank-decomposed matrices into the attention and feed-forward layers, allowing the model to adapt to new tasks by updating only a fraction of parameters. This approach reduces the number of trainable parameters by more than $10,000\times$ compared to full fine-tuning and maintains competitive performance in various natural language processing (NLP) tasks. The efficiency and simplicity of LoRA have made it a standard baseline for many subsequent PEFT studies.

Despite its success, LoRA applies the same adaptation strategy uniformly across all layers, implicitly assuming that all layers contribute equally to downstream tasks. This uniformity may overlook the varying importance of different layers, potentially leading to suboptimal performance or unnecessary computational overhead while it may not be necessary.

To address this limitation, methods like **Layerwise Importance Sampling for Memory-Efficient Large Language Model Fine-Tuning (LISA)** (Pan et al., 2024) selectively fine-tune layers based on calculated importance scores. LISA evaluates parameter gradient aggregation which is accumulating per-layer gradient magnitudes (e.g., L2-norm of gradients) or by estimating per-parameter variance. Layers with larger aggregate gradients are deemed more crucial since modifying them is expected to yield greater performance improvements. By identifying and updating only the most critical layers, LISA reduces computational costs and memory usage while preserving or even enhancing performance compared to full fine-tuning. Similarly, Yao et al. (Yao et al., 2024) propose a dynamic layer-wise analysis framework that adjusts which layers to fine-tune based on importance metrics, how constrained or expansive the effective parameter space of a layer is, and mutual information between layer representations and target outputs.

An alternative approach is **Layer-wise Fine-Tuning (LIFT)** (Zhu et al., 2024), which updates one layer at a time while freezing the rest. This method reduces the back-propagation depth, significantly improving memory efficiency and fine-tuning throughput compared to full fine-tuning. Unlike LoRA, which distributes learnable parameters across all layers, LIFT’s iterative optimization scheme focuses computational effort on a single layer at a time. However, LIFT’s layer-by-layer approach may introduce inefficiencies for tasks requiring coordinated updates across multiple layers and can result in slower convergence compared to methods that update multiple layers simultaneously.

While these approaches demonstrate effectiveness, they exhibit several notable limitations. First, they rely on empirical heuristics, such as gradient magnitudes or mutual information, without robust theoretical guarantees for their effectiveness. For instance, methods such as Importance-aware Sparse Tuning (IST) (Yao et al., 2024) effectively reduce the number of trainable parameters at each step of the training process. However, they do not provide theoretical guarantees that the total number of parameters trained throughout the process will be minimized. Furthermore, techniques like LISA and IST involve repeated computation of layer-wise metrics, which can become prohibitively expensive for large models with deep architectures, especially when scaling to multi-layer scenarios. For large models, these operations can scale unfavorably with model depth and parameter dimensionality, introducing an additional layer of complexity and latency. These limitations highlight the need for more systematic frameworks that combine the parameter efficiency of methods like LoRA with a deeper understanding of layer-wise contributions, enabling fine-tuning strategies that are both computationally efficient and effective across a wide range of tasks.

2.2. Layerwise Analysis and Attention Mechanisms

Layer-wise analysis has revealed that updates required for fine-tuning often reside in low-dimensional subspaces (Pan et al., 2024; Yao et al., 2024). This insight has spurred methods that exploit the low intrinsic dimensionality of model updates to enhance efficiency. However, understanding the nuanced role of each layer requires considering not just parameter importance but also how attention mechanisms operate within and across layers.

Attention mechanisms are fundamental to the success of transformer-based models, enabling them to weigh the relevance of different input tokens when generating representations. Recent work has identified the phenomenon of **attention sinks**, where certain tokens disproportionately attract attention without contributing meaningful semantic information (Yu et al., 2024). These attention sinks can hinder the model’s ability to focus on more informative tokens, potentially degrading performance.

The interplay between attention mechanisms and layer-wise contributions is crucial. Layers may differ in how they handle attention distributions, and attention sinks may affect some layers more than others. By integrating attention patterns into layer-wise analysis, we can gain a more comprehensive understanding of layer importance and attention sinks impact on layer efficiencies on processing information.

2.3. Attention Calibration Techniques and Their Limitations

To mitigate the impact of attention sinks, **Attention Calibration Technique (ACT)** (Yu et al., 2024) recalibrates attention distributions during inference by penalizing attention scores by focusing more on end tokens, reducing the influence of attention sinks. Although ACT improves task alignment and inference efficiency, it applies the same penalization to all tokens, potentially overlooking the varying semantic importance of different tokens. Moreover, ACT operates solely during inference and does not inform the fine-tuning process or consider how varying penalization might affect different layers.

In our work, we employ a **modified attention calibration technique** that employs a log-square-normalized position-based penalty to adjust attention scores, offering a more nuanced penalization compared to linear position-based methods. This approach applies gentler penalization to less informative tokens while preserving higher attention weights for more critical tokens. Using logarithmic decay, the technique ensures a smoother attenuation of attention, enabling the model to focus effectively on semantically relevant tokens which enhances overall performance.

2.4. Dynamic Loading and Efficiency of LoRA Adapters

Efficient switching and loading of adapter models are crucial for practical deployment, especially in resource-constrained environments. Apple highlighted their recent work on foundation models (Research, 2024). Adapter models like those used in LoRA can be dynamically loaded, temporarily cached in memory, and swapped. This capability allows a foundation model to specialize itself on the fly for the task at hand while efficiently managing memory and ensuring system responsiveness. Dynamically loading LoRA adapters facilitates quick adaptation to new tasks without the overhead of loading an entirely new task-specific model, making it particularly valuable for applications requiring rapid context switching.

2.5. Limitations and Open Challenges

While PEFT methods like LoRA, LISA, and the approach by Yao et al. have significantly advanced the field by reducing computational demands, they often rely on assumptions

that may not hold across different tasks or model architectures. For instance, assuming uniform layer importance or employing static importance metrics can lead to suboptimal fine-tuning strategies that waste computational resources or fail to achieve the best possible performance.

Attention-focused methods like ACT enhance inference by recalibrating attention distributions, but fail to integrate these insights into the fine-tuning process. They also do not consider how attention patterns interact with layer-wise dynamics and how this interplay affects the model’s adaptability to new tasks.

These limitations highlight a gap in existing methodologies: the lack of a systematic approach that combines attention calibration with dynamic layer-wise analysis to guide fine-tuning decisions. An ideal method would consider the varying contributions of different layers in the context of attention mechanisms, employing attention calibration techniques that apply variable penalization to attention scores for more targeted adjustments. Additionally, it would leverage the dynamic loading of LoRA adapters to enable efficient specialization and resource management.

2.6. Contributions of Our Work

Our work makes the following contributions:

- **C-SAF Algorithm:** We introduce a novel method to find **Non-Complementary** (A_{nc}) and **Complementary** (A_c) attention layers for a transformer based Large Language Model (LLM) L for a Task t . We show that only $n = 25\%$ of L , non-complementary layers are sufficient for fine-tuning.
- **Selective Fine-Tuning:** We show that selectively fine-tuning non-complementary layers, as guided by C-SAF, achieves superior performance compared to existing PEFT methods, while significantly reducing computational overhead.
- **Dynamic Loading of Adapters:** We show that we can reduce the LoRA adapter size by **75%** while improving the accuracy in downstream tasks. Decreased adapter size is particularly important for dynamically swapping adapters in resource constrained environments as highlighted by Apple (Research, 2024)
- **Extensive Experimental Validation:** We validate our approach through extensive experiments on benchmark NLP tasks and various SOTA open source models, demonstrating its effectiveness and generalizability.

By addressing the interplay between attention mechanisms and layer-wise contributions, and by leveraging the dynamic capabilities of adapter models, our work offers a significant advancement in the field of efficient model fine-tuning.

3. Motivation

Attention Calibration Technique (ACT): The Attention Calibration Technique (ACT) reveals that Attention Sinks play a critical role in contextual learning within LLMs. By redistributing attention weights to mitigate imbalances, ACT offers a lens into the mechanics of attention patterns. This exploration holds the potential to uncover deeper insights into task-specific attention mechanisms and their impact on model performance, shedding light on how LLMs process and prioritize contextual information.

Selective Fine-Tuning with LoRA Adapters: Motivated by the findings of SubTuning (Kaplun et al., 2023) there exists a set of layers when fine-tuned can match or surpass the performance of full fine-tuning and also lead to a more generalized model. We hypothesize that we can use ACT to find these set of layers and only fine-tuning these layers using PEFT LoRA will lead to a more efficient model which is able to match the accuracy of full-fine-tuning.

To bridge these gaps, we propose **Calibration-Driven Sparse Attention Fine-Tuning (C-SAF)**, a novel framework that utilizes attention calibration technique to find most optimal target n target layers to fine-tune. Our approach (Figure 1) operates in two key phases :

Phase I : Attention Calibration

We employ a modified ACT (Yu et al., 2024) that utilizes log-square normalization for redistributing attention scores, ensuring consistent penalization based on position across tokens.

- First we sample S as $\max(4000, 40\% * D)$ where D is the Training set of Task t .
- For each layer i in Large Language Model M . We evaluate the model M on S while calibrating layer i as depicted in Algorithm 1.
- We record the accuracy for calibration on each i th layer. Sort the layers by accuracy and define any calibrated layer i which leads to accuracy increase compared to original model on set S for model M as non-complementary, otherwise complementary.

By analyzing changes in task accuracy resulting from calibration across different layers, we assess "are the current attention weights in layer i complementing the task t ?". As illustrated in Figures 2 and 3, our $\log^2$ position-based normalization approach provides more nuanced attention score adjustments compared to linear normalization, enabling better identification of truly non-complementary layers. Since attention sinks hold very significant attention

Algorithm 1 Attention Sink Detection and Layer Classification

Require: Model M with L layers, hyperparameter β , evaluation dataset D

Ensure: Sets of complementary and non-complementary layers

```

 $S \leftarrow \min(4000, 0.4 \times |D|)$ 
baseline_acc  $\leftarrow$  Evaluate( $M, S$ )
layer_impacts  $\leftarrow$  []
for  $l = 1, \dots, L$  do
     $M_l \leftarrow$  Clone( $M$ )
    for each attention head  $h$  in layer  $l$  do
        {Detect attention sinks}
         $S_h^l \leftarrow$  DetectSinks( $A_h^l$ )
        {Reduce attention scores for sinks}
        for each row  $k$  of  $A_h^l$  do
            for each  $s \in S_h^l$  do
                 $\hat{A}_h^l[k, s] \leftarrow A_h^l[k, s] \times \beta$ 
            end for
        end for
        {Redistribute attention to non-sink tokens}
        for each row  $k$  of  $A_h^l$  do
            for each  $s \notin S_h^l$  do
                 $\Delta_{ks} \leftarrow \sum_{u \in S_h^l} (\hat{A}_h^l[k, u] - A_h^l[k, u])$ 
                 $w_t \leftarrow \frac{A_h^l[k, t]}{\sum_{i \notin S_h^l} A_h^l[k, i]}$ 
                 $\hat{A}_h^l[k, s] \leftarrow A_h^l[k, t] + \Delta_{ks} \times w_t$ 
            end for
        end for
    end for
    {Evaluate modified layer}
    modified_acc  $\leftarrow$  Evaluate( $M_l, S$ )
    impact  $\leftarrow$  modified_acc - baseline_acc
    layer_impacts.append(( $l$ , impact))
end for
{Sort layers by impact}
sorted_layers  $\leftarrow$  Sort(layer_impacts, key = impact, descending = True)
{Classify layers}
non_complementary  $\leftarrow$  { $l \mid (l, \text{impact}) \in \text{sorted\_layers} \wedge \text{impact} > 0$ }
complementary  $\leftarrow$  { $l \mid (l, \text{impact}) \in \text{sorted\_layers} \wedge \text{impact} \leq 0$ }
return (non_complementary, complementary)

```

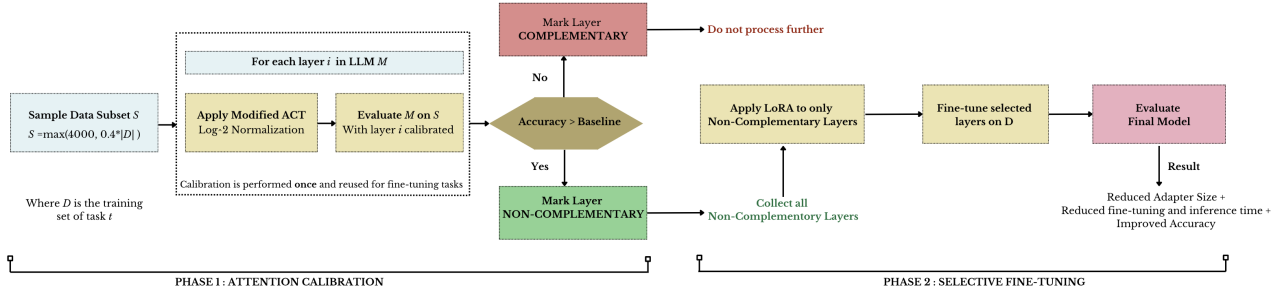


Figure 1. Workflow of the Calibration-Driven Sparse Attention Fine-Tuning (C-SAF) methodology. **Phase I** identifies non-complementary layers through attention calibration, a one-time task, while **Phase II** fine-tunes these layers selectively for optimal performance.

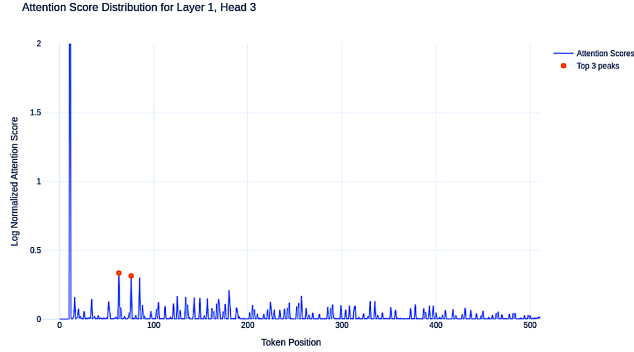


Figure 2. (C-SAF) $\log^2$ position based normalization of attention scores

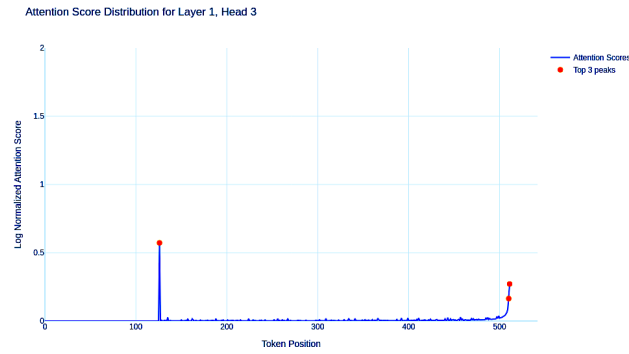


Figure 3. (Original ACT) Linear position based normalization of attention scores

score, if redistributing them is increasing the accuracy then we hypothesize that the current attention weights are non-complementary. This approach enables a deeper understanding of task-specific attention mechanisms, their influence on model performance, and how LLMs process and prioritize contextual information.

Phase II : Selective Fine-Tuning

Based on the calibration analysis, we identify non-complementary layers, layers where calibration alone does not sufficiently improve performance. We then selectively fine-tune the *non-complementary* layers.

By using attention calibration to measuring to layer importance for fine-tuning, C-SAF addresses the shortcomings of existing PEFT methods that assume uniform layer contributions or rely on static importance metrics. Our approach leverages dynamic attention patterns to inform fine-tuning strategies, leading to optimal task performance with minimal computational overhead.

Building on the findings of SubTuning (Kaplun et al., 2023), which demonstrated that fine-tuning a specific subset of layers can achieve performance comparable to or even exceeding that of full fine-tuning while also promoting greater model generalization, we propose a novel approach. We hypothesize that our modified ACT can effectively identify these critical layers. By selectively fine-tuning only these layers using parameter-efficient methods such as LoRA, we aim to develop a more efficient model that retains the accuracy of full fine-tuning while significantly reducing computational and memory overhead.

4. Experimental Results

4.1. Evaluation settings

Our results from Tables 1 and 2 further demonstrate the ability of C-SAF to meet the dual demands of efficiency and accuracy in modern AI systems by optimizing GPU

Table 1. Accuracy (%) on Multiple Choice Datasets

Model	Method	Hellaswag	ARCE	ARCC	PIQA	OB	MMLU	COPA	CQA	Avg.
Llama3	Baseline	92.65	91.58	82.27	87.49	86.20	58.85	98.00	81.98	84.87
	C-SAF	93.54	91.75	81.94	89.72	88.00	61.71	98.00	83.70	86.05
Mistral	Baseline	93.16	91.58	80.27	87.49	87.80	58.02	98.00	83.62	84.99
	C-SAF	93.66	91.93	79.93	89.34	87.20	59.29	96.00	83.70	?

Table 2. Finetuning Time (minutes) on Multiple Choice Datasets

Model	Method	Hellaswag	ARCE	ARCC	PIQA	OB	MMLU	COPA	CQA	Avg.
Llama3	Baseline	37.13	8.65	3.12	37.88	18.67	37.68	1.55	36.87	22.69
	C-SAF	25.63	6.38	2.75	28.12	13.75	27.98	1.08	27.02	16.59
Mistral	Baseline	36.60	8.35	4.25	36.47	18.10	36.85	1.57	35.55	22.22
	C-SAF	27.05	5.50	2.83	26.22	13.60	27.29	1.35	26.07	16.23

resources without compromising on performance. We use the baseline as fine-tuning all layers of the model using LoRA.

As shown in Figure 4, the integration of C-SAF significantly reduces the LoRA adapter size on Llama3 8B from 216 MB to 54 MB, achieving an impressive 75% reduction while maintaining model performance. This substantial decrease in adapter size emphasizes the practicality of C-SAF in deployment scenarios, particularly in resource-constrained environments. By minimizing resource requirements without compromising accuracy, as we show in Table 1, C-SAF broadens the applicability of fine-tuning large language models across diverse hardware settings.

We evaluate C-SAF on two state-of-the-art models—Llama3.1-8B (Dubey et al., 2024) and Mistral-Instruct-v0.3-7B (Jiang et al., 2023)—to further highlight its efficiency. As visualized in Figure 5, our experimental results show that C-SAF achieves a substantial reduction in GPU memory usage during fine-tuning, stabilizing at 70% (60.31GB) of total H100 GPU memory compared to the baseline’s 90% (75GB). This corresponds to a 14.69GB memory saving (approximately 20% reduction).

As illustrated in Figure 6, the convergence behavior on the Hellaswag dataset highlights the efficiency of our selective layer fine-tuning strategy. The C-SAF approach, employing only 25% layers, attains more rapid convergence than the 32-layer baseline, thereby validating the practicality and effectiveness of C-SAF. This, along with our results from Table 2, show the efficiency of selective layer fine-tuning in accelerating training convergence for large language models.

Figures 7 and 8 provides visual evidence of our layer classi-

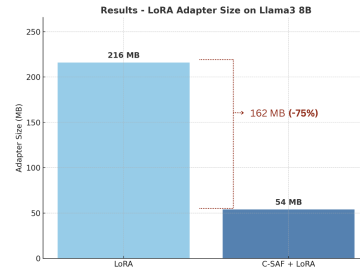


Figure 4. Comparison of LoRA Adapter Sizes demonstrating the resource efficiency of C-SAF during fine-tuning. The integration of C-SAF reduces the adapter size on Llama3 8B from 216 MB to 54 MB, achieving a 75% reduction while preserving model performance.

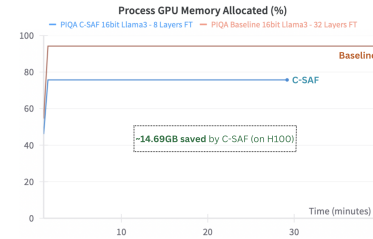


Figure 5. Process GPU Memory Allocated (%) during fine-tuning on the PIQA dataset using an H100 GPU (80GB capacity). The C-SAF method (16bit Llama3, 8 Layers) stabilizes at 70% (60.31GB) of total GPU memory, compared to the Baseline method (16bit Llama3, 32 Layers) stabilizing at 90% (75GB).

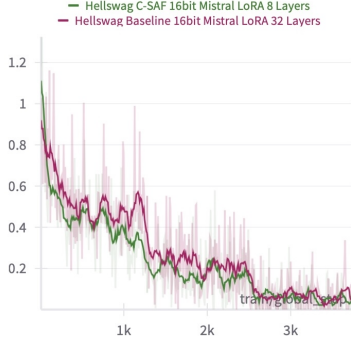


Figure 6. Running Average Loss on the Hellaswag dataset. The C-SAF approach (16bit Mistral LoRA, 8 layers) converges more rapidly than the baseline (16bit Mistral LoRA, 32 layers).

fication approach, contrasting attention maps between complementing and non-complementing layers. While complementing layers show minimal changes post-calibration despite accuracy improvements, non-complementing layers exhibit significant attention pattern shifts that correlate with performance gains, supporting our layer selection methodology.

We choose minimum of 25% layers of the model and number of non-complementary layers. This way we make sure we are reducing the number of training parameters by at least 75%. We only finetune 25% layers based on empirical findings. First, we consistently see across some widely known benchmarks that the number of non-complementary layers come around 8, See Table 3. Next we run experiments fine-tuning different number of layers and the gains peak at 20-25% range. Table 4.

Task	No. of non - complementary layers
Hellaswag	8
MMLU	8
ARCC	8
ARCE	9

Table 3. Number of complementary layers for various tasks on llama3.1

Task	5L	6L	7L	8L
MMLU	62.33	62.83	62.48	61.71
Hellaswag	93.45	93.51	93.51	93.54

Table 4. Performance metrics for different layers (L) across tasks

5. Conclusion

In this paper, we introduced **Calibration-Driven Sparse Attention Fine-Tuning (C-SAF)**, a novel framework that

leverages attention calibration to systematically identify non-complementary layers for selective fine-tuning. By recognizing and focusing on the subset of layers most critical to downstream performance, C-SAF achieves substantial latency and memory savings while enhancing accuracy. Our approach integrates insights from attention calibration with dynamic layer selection, offering a more refined alternative to uniform PEFT strategies such as LoRA. Experiments across multiple large language models and tasks demonstrate that C-SAF not only reduces the LoRA adapter size by over 75% but also achieves notable improvements in fine-tuning and inference latency. These results underscore the immense potential of our methodology to streamline LLM adaptation in both large-scale data centers and on-device scenarios, where resource constraints are paramount. Based on observations that selective layer tuning can outperform full fine-tuning, C-SAF provides a new, more agile paradigm for LLM fine-tuning and deployment, one that stands as a compelling successor to existing PEFT solutions.

Software and Data

The code is available at <https://github.com/gatech/kshaikh6/auto-act/>

Impact Statement

C-SAF’s ability to reduce computational resources while maintaining or improving model performance has several key domain applications. Firstly, we think it can democratize access to LLM fine-tuning capabilities for researchers and organizations with limited compute. C-SAF could also accelerate specialized language model deployment across various resource-constrained domains which may include healthcare, education, and scientific research. We strongly believe the on-device applications will be privacy-preserving and reduce the need for cloud-based processing. However with a lower barrier to entry, this could cause the proliferation of specialized language models and may spread harmful applications. Some techniques to combat this could include robust validation procedures, adding in-built model guardrails, or even encouraging transparency in applications. We believe our findings with C-SAF could influence future directions in parameter-efficient fine-tuning methods and further model compression and distillation techniques through identifying both inter-layer and intra-layer relationships.

References

Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., Goyal, A., Hartshorn, A., Yang, A., Mitra, A., Sravankumar, A., Korenev, A., Hinsvark, A., Rao, A., Zhang, A.,

ARCE Dataset, Sample 19, Layer 5 (most important), Head 0

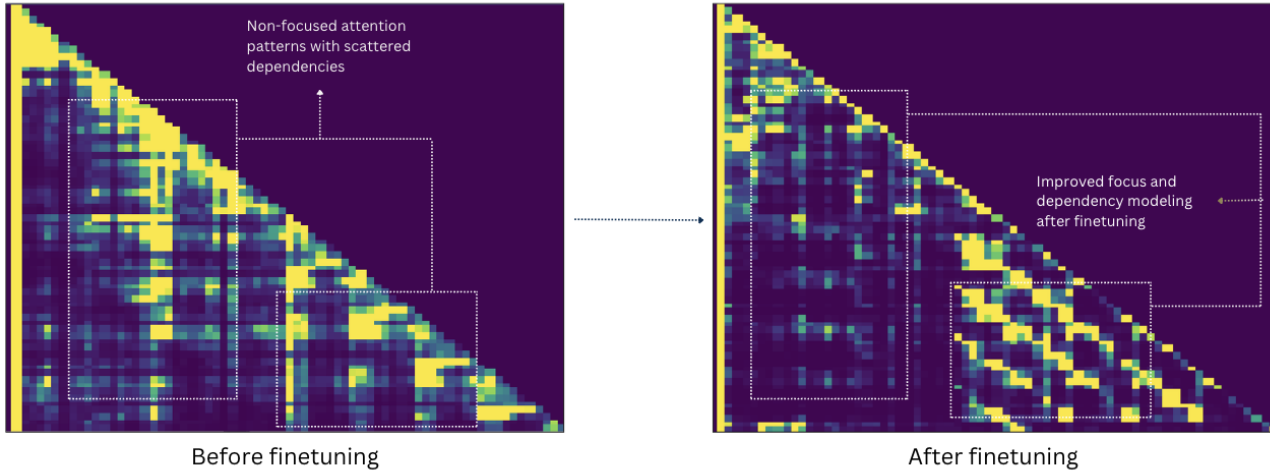


Figure 7. Attention map for a non complementing layer on a sample that improves accuracy after calibration driven finetuning, with significant changes in the maps leading to the correction.

ARCE Dataset, Sample 19, Layer 31 (least Important), Head 0

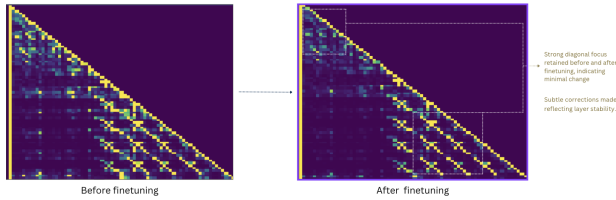


Figure 8. Attention map for a complementing layer on a sample that improves accuracy after calibration driven finetuning, though the maps show minimal change despite the correction.

Rodriguez, A., Gregerson, A., Spataru, A., Rozière, B., Biron, B., Tang, B., Chern, B., Caucheteux, C., Nayak, C., Bi, C., Marra, C., McConnell, C., Keller, C., Touret, C., Wu, C., Wong, C., Ferrer, C. C., Nikolaidis, C., Al-lonsius, D., Song, D., Pintz, D., Livshits, D., Esiobu, D., Choudhary, D., Mahajan, D., Garcia-Olano, D., Perino, D., Hupkes, D., Lakomkin, E., AlBadawy, E., Lobanova, E., Dinan, E., Smith, E. M., Radenovic, F., Zhang, F., Synnaeve, G., Lee, G., Anderson, G. L., Nail, G., Mialon, G., Pang, G., Cucurell, G., Nguyen, H., Korevaar, H., Xu, H., Touvron, H., Zarov, I., Ibarra, I. A., Kloumann, I. M., Misra, I., Evtimov, I., Copet, J., Lee, J., Geffert, J., Vranes, J., Park, J., Mahadeokar, J., Shah, J., van der Linde, J., Billock, J., Hong, J., Lee, J., Fu, J., Chi, J., Huang, J., Liu, J., Wang, J., Yu, J., Bitton, J., Spisak, J., Park, J., Rocca, J., Johnstun, J., Saxe, J., Jia, J., Alwala, K. V., Upasani, K., Plawiak, K., Li, K., Heafield, K., Stone, K., and van der Maaten, L. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., et al. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. URL <https://arxiv.org/abs/2106.09685>.

Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Renard Lavaud, L., Lachaux, M.-A., Stock, P., Le Scao, T., Lavril, T., Wang, T., Lacroix, T., and El Sayed, W. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023. URL <https://arxiv.org/abs/2310.06825>.

Kaplun, G., Gurevich, A., Swisa, T., David, M., Shalev-Shwartz, S., and Malach, E. Less is more: Selective layer finetuning with subtuning, 2023. URL <https://arxiv.org/abs/2302.06354>.

Pan, R., Liu, X., Diao, S., Pi, R., Zhang, J., Han, C., and Zhang, T. Lisa: Layerwise importance sampling for memory-efficient large language model fine-tuning. *arXiv preprint arXiv:2403.17919*, 2024. URL <https://arxiv.org/abs/2403.17919>.

Research, A. M. L. Introducing apple’s on-device and server foundation models, 2024. URL <https://machinelearning.apple.com/research/introducing-apple-foundation-models>.

Yao, K., Gao, P., Li, L., Zhao, Y., Wang, X., Wang, W., and Zhu, J. Layer-wise importance matters: Less memory for better performance in parameter-efficient fine-tuning of large language models. *arXiv preprint*

arXiv:2410.11772, 2024. URL <https://arxiv.org/abs/2410.11772>.

Yu, Z., Wang, Z., Fu, Y., Shi, H., Shaikh, K., and Lin, Y. C. Unveiling and harnessing hidden attention sinks: Enhancing large language models without training through attention calibration. *arXiv preprint arXiv:2406.15765*, 2024. URL <https://arxiv.org/abs/2406.15765>.

Zhu, L., Hu, L., Lin, J., and Han, S. Lift: Efficient layer-wise fine-tuning for large language models. In *International Conference on Learning Representations (ICLR)*, 2024. <https://openreview.net/forum?id=u0INlprg3U>.

Hyperparameters (LoRA + IST)	Commonsense Reasoning	
	Mistral-Instruct-v0.3 _{7B}	LLaMA _{38B}
Rank r	32	32
Dropout	0.05	0.05
Optimizer	AdamW	AdamW
LR	2e-4	1e-4
LR Scheduler	Warmup Steps	Warmup Steps
Batch size	8	8
Warmup Steps	100	100
Epochs	3	3
Where	{Q, K, V, Up, Down}	{Q, K, V, Up, Down}
μ	3	3
β	0.1	0.1
s (% sampled training data for layer classification)	40%	40%
n (% of total model layers fine-tuned)	25%	25%

Table 5. Hyperparameters used for LoRA + CSAF fine-tuning across models and tasks.